WORKED SOLUTION

STEP III 1998, Question 2

Pure • Beta Integral Recurrences

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q2

The key idea

This integral is the beta function in disguise. Each part needs one standard move. Part (i) uses the reflection $t \mapsto 1-t$, which swaps the roles of t and $1-t$. Part (ii) multiplies the integrand by $1 = t + (1-t)$. Part (iii) integrates by parts, which moves one power from $(1-t)$ onto t. Repeating part (iii) until the power of $(1-t)$ is zero leaves an integral we can do at once.

Solution

Part (i)

Substitute $u = 1-t$, so $\mathrm{d}t = -\mathrm{d}u$, and the limits $t = 0, 1$ become $u = 1, 0$:

$$\mathrm{I}(a,b) = \int_1^0 (1-u)^a u^b \,(-\mathrm{d}u) = \int_0^1 u^b (1-u)^a \,\mathrm{d}u = \mathrm{I}(b,a).$$

The minus sign from $\mathrm{d}t = -\mathrm{d}u$ is used up in swapping the limits back to 0 and 1.

Part (ii)

Since $t + (1-t) = 1$, we can write the integrand as

$$t^a(1-t)^b = t^a(1-t)^b(t+(1-t)) = t^{a+1}(1-t)^b + t^a(1-t)^{b+1}.$$

Integrating each term over $[0,1]$ gives $\mathrm{I}(a,b) = \mathrm{I}(a+1,b) + \mathrm{I}(a,b+1)$, as required.

Part (iii)

We integrate $\mathrm{I}(a,b)$ by parts, differentiating $(1-t)^b$ and integrating t^a:

$$\begin{aligned}\mathrm{I}(a,b) &= \left[\frac{t^{a+1}}{a+1}(1-t)^b\right]_0^1 - \int_0^1 \frac{t^{a+1}}{a+1}\cdot(-b(1-t)^{b-1})\,\mathrm{d}t \\ &= \left[\frac{t^{a+1}}{a+1}(1-t)^b\right]_0^1 + \frac{b}{a+1}\int_0^1 t^{a+1}(1-t)^{b-1}\,\mathrm{d}t.\end{aligned}$$

> **Ask yourself** Why does the boundary term vanish, and where do we need a and b to be positive?

At $t=1$ the factor $(1-t)^b$ is 0 because $b>0$. At $t=0$ the factor t^{a+1} is 0 because $a+1>0$. So the bracket is zero, and multiplying through by $a+1$ gives

$$(a+1)\,\mathrm{I}(a,b) = b\,\mathrm{I}(a+1,b-1),$$

which is the result. (If $0<b<1$, the last integral has a negative power of $(1-t)$. It still converges, because $b-1>-1$. For the integer case below we always have $b-1\geq 0$ anyway.)

Calculating $\mathrm{I}(a,b)$ for integer a,b. Write the result as $\mathrm{I}(a,b) = \dfrac{b}{a+1}\,\mathrm{I}(a+1,b-1)$ and apply it again and again. Each step raises the first argument by 1 and lowers the second by 1. We stop once the second argument reaches 0:

$$\begin{aligned}\mathrm{I}(a,b) &= \frac{b}{a+1}\,\mathrm{I}(a+1,b-1) = \frac{b}{a+1}\cdot\frac{b-1}{a+2}\,\mathrm{I}(a+2,b-2) = \cdots \\ &= \frac{b(b-1)\cdots 1}{(a+1)(a+2)\cdots(a+b)}\,\mathrm{I}(a+b,0).\end{aligned}$$

Every step is legitimate, since the second argument is a positive integer each time we use the formula. Now $\mathrm{I}(a+b,0) = \int_0^1 t^{a+b}\,\mathrm{d}t = \dfrac{1}{a+b+1}$. Also $(a+1)(a+2)\cdots(a+b) = \dfrac{(a+b)!}{a!}$, so

$$\boxed{\mathrm{I}(a,b) = \frac{a!\,b!}{(a+b+1)!}}$$

The formula is symmetric in a and b, as part (i) says it must be. It also holds when a or b is 0.

Checks

- $\mathrm{I}(1,1) = \int_0^1 (t-t^2)\,\mathrm{d}t = \frac{1}{2}-\frac{1}{3} = \frac{1}{6}$, and the formula gives $\frac{1!\,1!}{3!} = \frac{1}{6}$.
- The formula satisfies part (ii): $\frac{(a+1)!\,b!}{(a+b+2)!} + \frac{a!\,(b+1)!}{(a+b+2)!} = \frac{a!\,b!\,(a+b+2)}{(a+b+2)!} = \frac{a!\,b!}{(a+b+1)!}$.

- Numerically, Simpson's rule gives $\mathrm{I}(2,3) \approx 0.0166667$, which matches $\frac{2!\,3!}{6!} = \frac{1}{60}$. We also checked parts (i)–(iii) by quadrature at random non-integer a, b, with agreement to about 10^{-11}.

Common slips

- Integrating by parts the other way round. Differentiating t^a raises the power of $(1-t)$, which is the wrong direction.
- Forgetting to justify the boundary term. It vanishes only because $a+1 > 0$ and $b > 0$.
- Stopping the recursion one step early, or forgetting the final factor $\frac{1}{a+b+1}$ from $\mathrm{I}(a+b, 0)$.

Beyond STEP

$\mathrm{I}(a,b)$ is the beta function $B(a+1, b+1)$. For all real positive arguments it is linked to the gamma function by $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$. Our answer is the special case $\Gamma(n+1) = n!$. In probability, $t^a(1-t)^b/\mathrm{I}(a,b)$ is the density of the beta distribution, which is central in Bayesian statistics. Blitzstein and Hwang, *Introduction to Probability*, has an attractive treatment, including a counting argument that explains why the answer is $a!\,b!/(a+b+1)!$.

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